ILLUnatic:

Effective Bot Screening on Web with Optical Illusion

[CSE551] Advanced Computer Security

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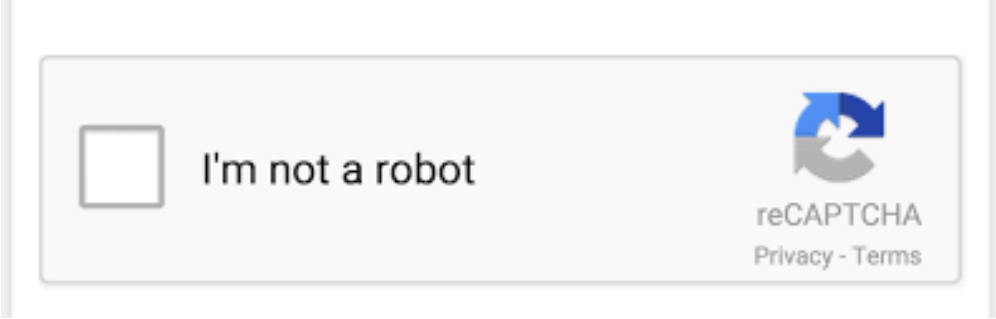
Introduction

- CAPTCHA
  - **C**ompletely **A**utomated **P**ublic **T**uring test to tell **C**omputers and **H**umans **A**part
  - An automated, public test to determine whether a website visitor is a human or a bot



Introduction

- 3 pain points of current CAPTCHA
  - High-performance OCR, image recognition easily pass character/object recognition-based CAPTCHAs
  - Increasingly complex and difficult testing harms the user experience
  - Modern systems like reCAPTCHA v3 collect and analyze user behavioral data, but the evaluation process is not transparent and raises privacy concerns

  - Especially, since advanced multimodal AI can recognize objects in complex images, understand context, and answer questions, existing image-based CAPTCHAs are no longer an effective defense

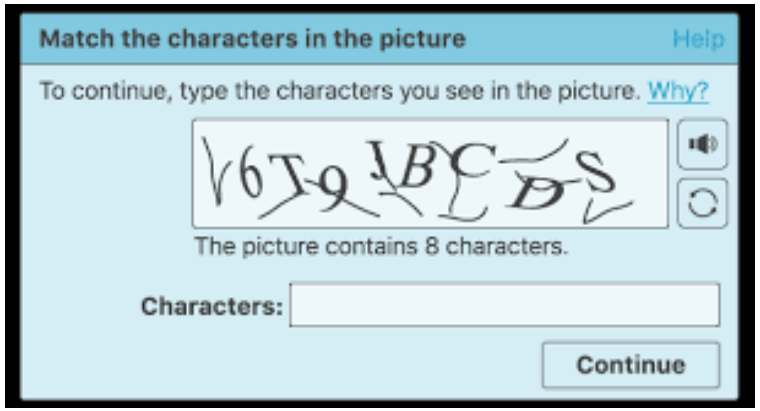
Introduction

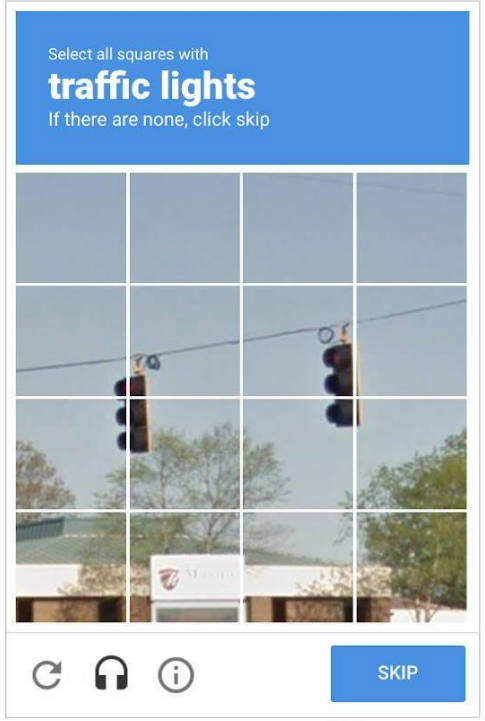
- We introduce **ILLUnatic:**
  - A CAPTCHA system that aims to clearly distinguish between normal users and bots by exploiting optical illusions that arise from differences in how humans and bots process visual information
  - Normal users can easily pass through it due to their unique visual information processing characteristics, but AI bots that rely entirely on image data are expected to be effectively filtered out

  - In this work, we evaluated several variables to determine how the illusionary images generated by ILLUnatic behave differently for humans and AI

Background: CAPTCHA

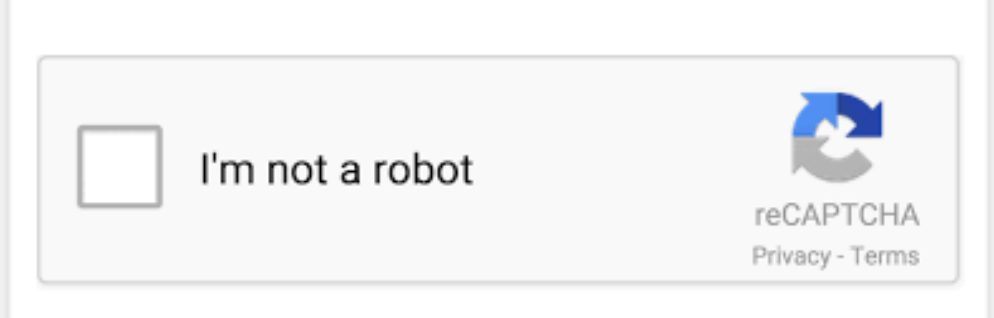
- 1st generation of CAPTCHA: Character-based
  - Recognize and input distorted or noisy letters/numbers
  - At the time, the computer's OCR technology was significantly inferior to human recognition capabilities
  - Advances in OCR technology have made machines more capable of reading text, rendering CAPTCHA meaningless
- 2nd generation of CAPTCHA: Image-based
  - Select an image containing a specific object or enter a word presented as audio
  - Lack of universality, such as using images more familiar to specific cultures
  - And of course, AI

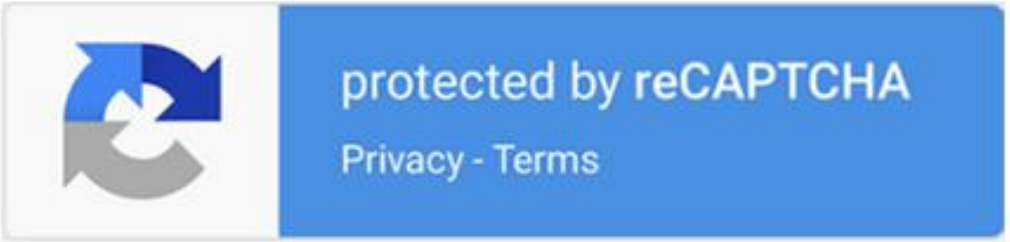




Background: CAPTCHA

- 3rd generation of CAPTCHA: Behavior-based
  - Analyzes user mouse movements (reCAPTCHA v2) or interactions with web pages (v3)
  - While the user experience has improved, concerns have been raised about privacy violations due to the collection and analysis of behavioral data

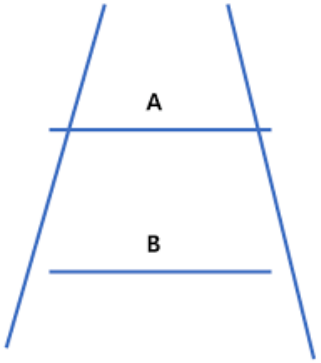




Background: Optical Illusion

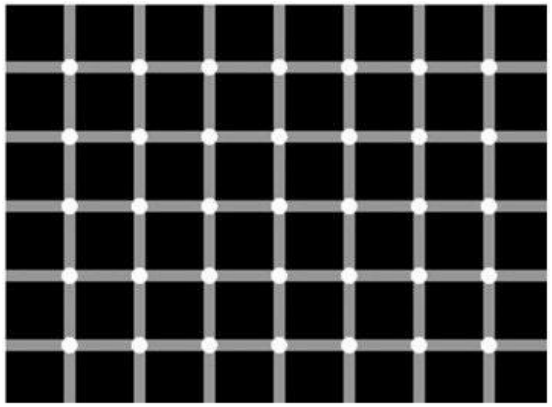
- 3 types of optical illusions
  1. Literal optical illusions
     - Occurs due to the physical properties of the substance itself or its interaction with the surrounding environment

  2. Physiological illusions
     - Caused by eye fatigue or excessive stimulation
  3. **Cognitive illusions**
     - **Occurs in the process of the brain inferring and interpreting based on existing knowledge or experience**



Background: Optical Illusion

- Hermann Grid Illusion
  - There are white dots at the intersections of the grids, but when human don't look directly at the intersections of the grids, it looks as if a blurry gray or black dot appears and disappears at the intersections
  - This illusion induced by lateral inhibition, which is part of the visual system so there is no difference in the occurrence of illusions depending on the user's age or culture sphere



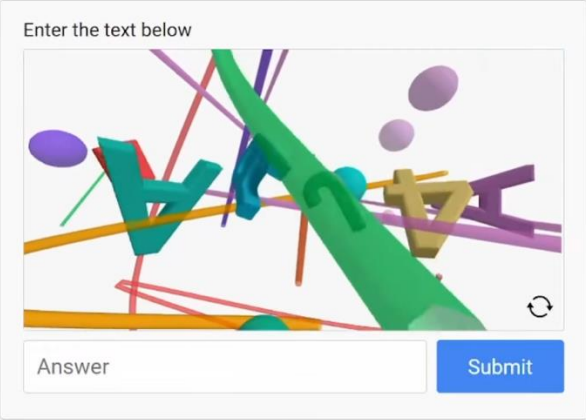
Related Works

- IllusionCAPTCHA [WWW 25]
  - ➢ Presents a specific methodology for transforming a specific image into another image containing an optical illusion using the 'Illusion Diffusion Model’

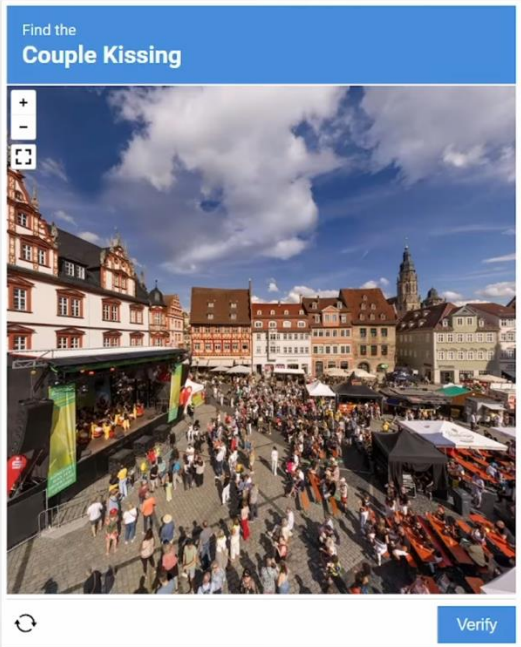
Related Works

- I’m Not a Robot [NEAL.FUN]
  - ➢ A Web-based game which contains several weird CAPTCHA scenarios
  - ➢ Although it is not a research work, it gives few reasonable CAPTCHA ideas that are worth to study

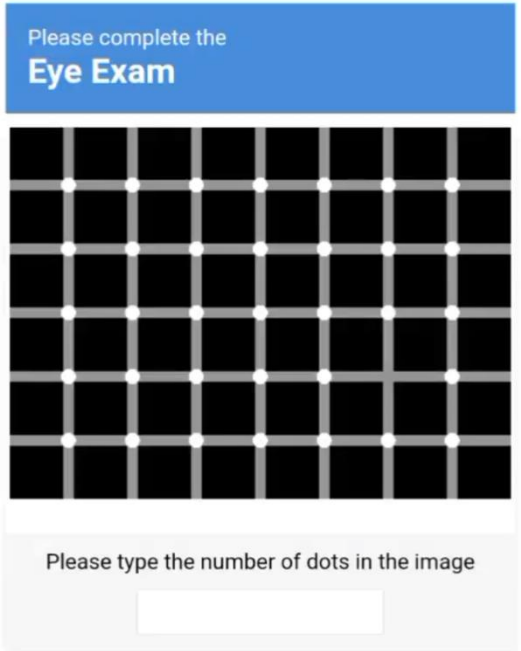
Word finding



3D characters



Panorama images



Optical Illusion

Design

- Main idea
  - “Accuracy alone isn't enough to distinguish AI. We need to look at **execution time**.”

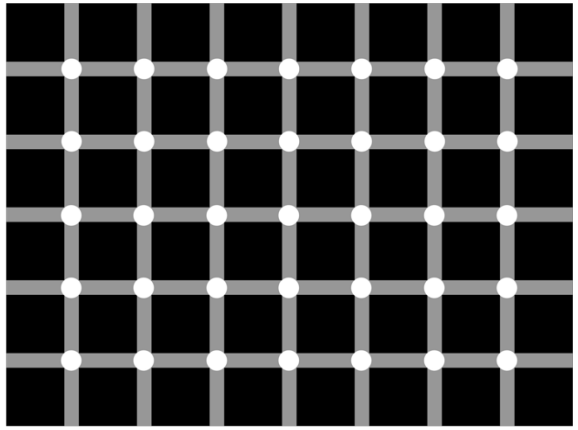
- Even if the biological mechanisms that cause optical illusions are the same, the strength of the illusion varies depending on the characteristics of the image
  - e.g., distance between intersections, contrast of colors

- Hypothesis: Illusionary CAPTCHAs with different image characteristics will produce different performance time tendencies between humans and AI

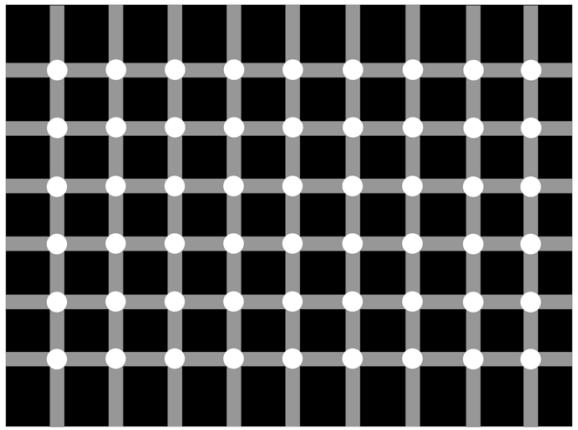
Evaluation

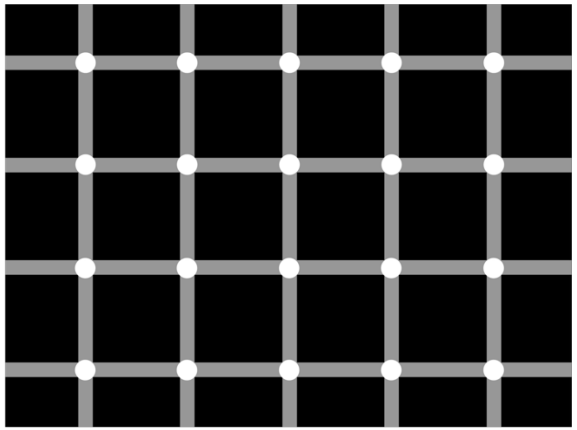
- Characteristics of the generated images
  - 4:3 aspect ratio, fixed background color (black)
  - White circles are randomly placed at ~80% of the intersections

  - Images from each category has same image property, except for one factor
    - In ‘Density’ category, the distance between intersections can be vary
    - In ‘Shape’ category, the shape of the intersecting lines changes
    - In ‘Brightness’ category, the brightness of the line changes

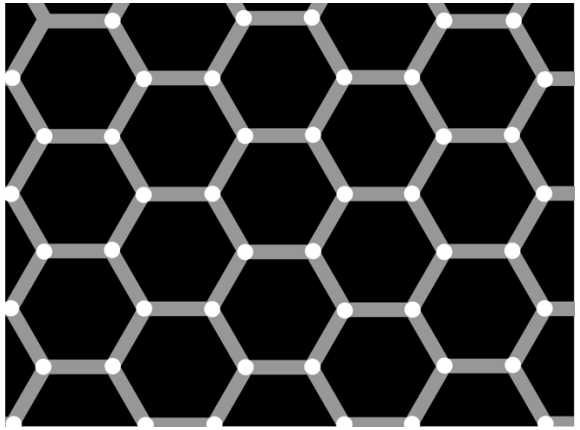


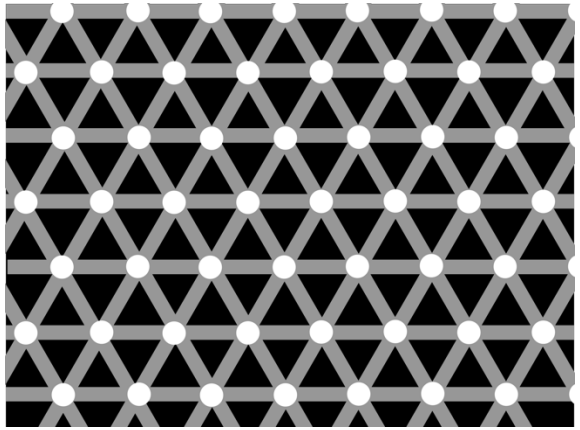
Density



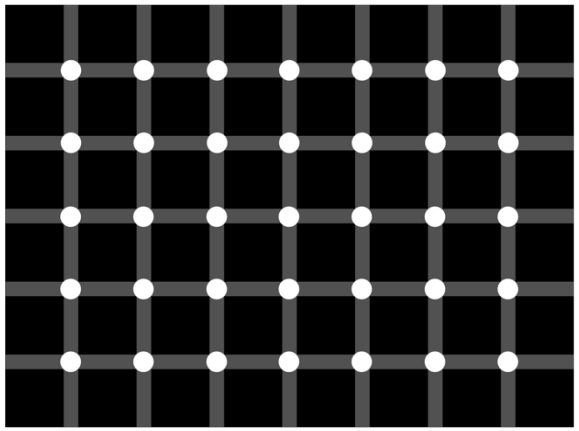


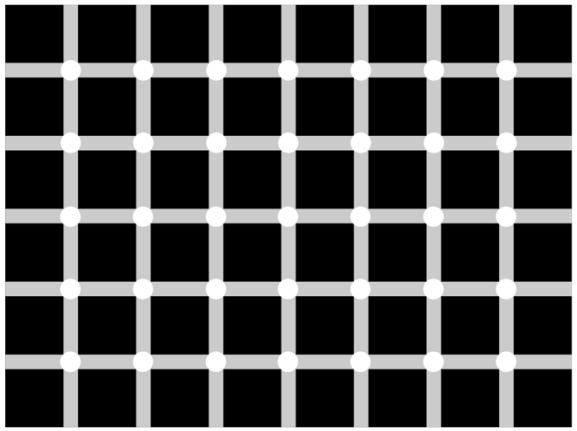
Shape





Brightness

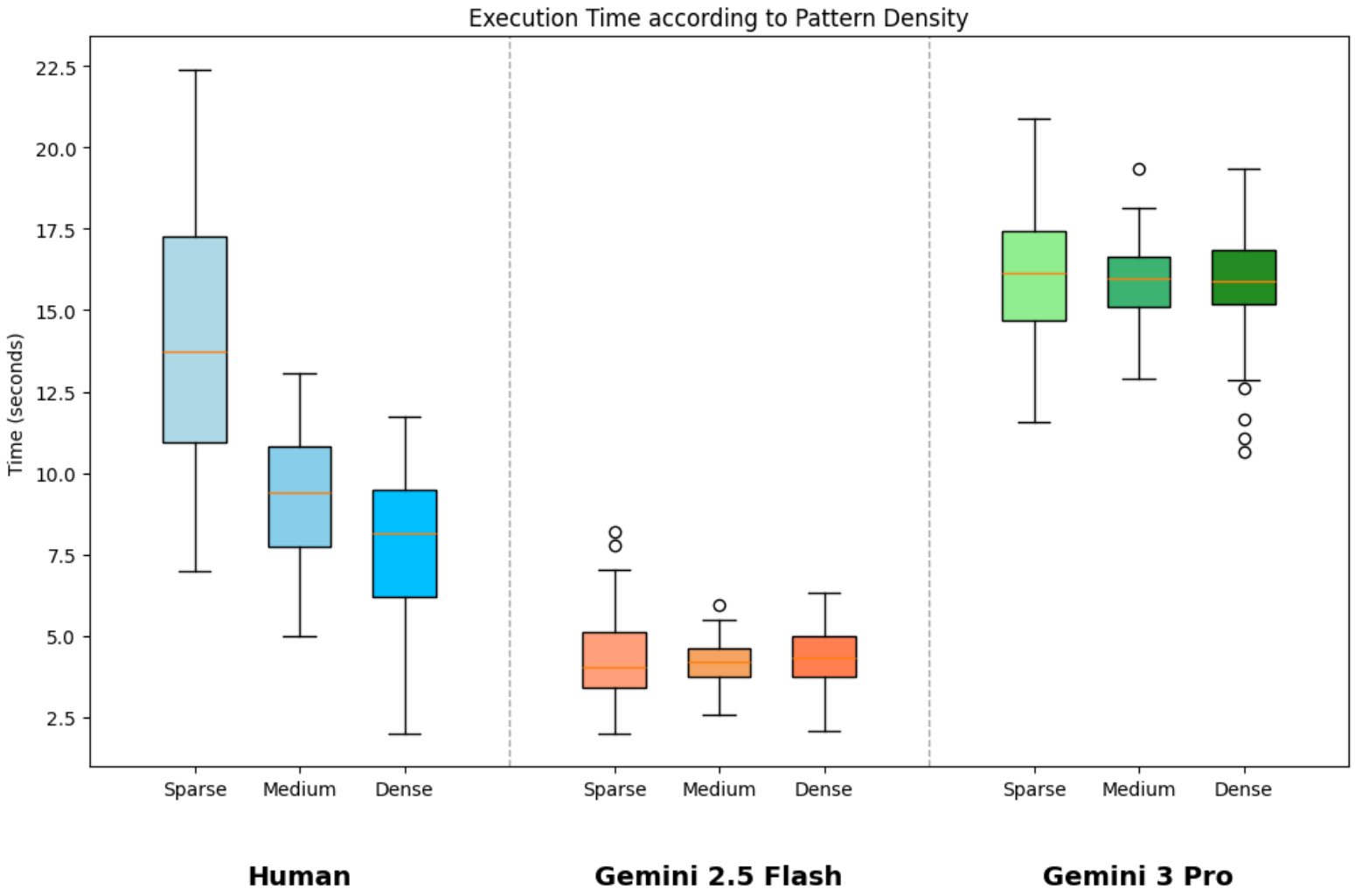




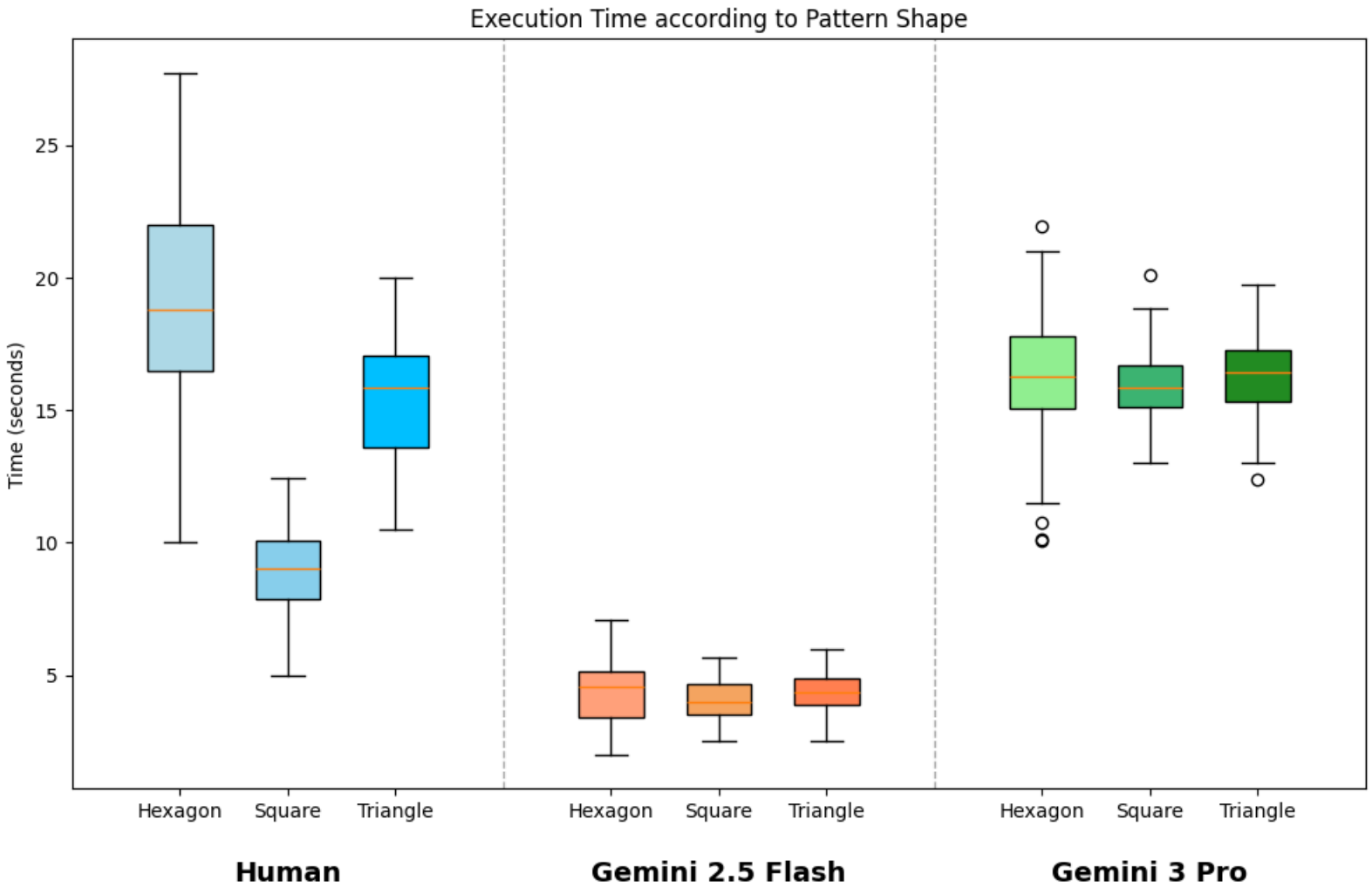
Evaluation

- Method
  - 12 Participants (4F 8M, 20.8 y.o. average) and 2 AI models (Gemini 2.5 Flash / 3 Pro)
  - Participants were instructed to count the number of white circles in the image as accurately as possible
  - Images from ILLUnatic shows on monitor, ~60cm between participants and monitors
  - 15 images of each of the three categories are displayed in the same size
    - Density (Sparse | Medium | Dense)
    - Shape (Hexagon | Square | Triangle)
    - Brightness (Dark | Medium | Bright)
    - In total, 12x5x9 = 540 images were examined with human participants
  - Measure the execution time it takes to arrive at the correct answer for each image

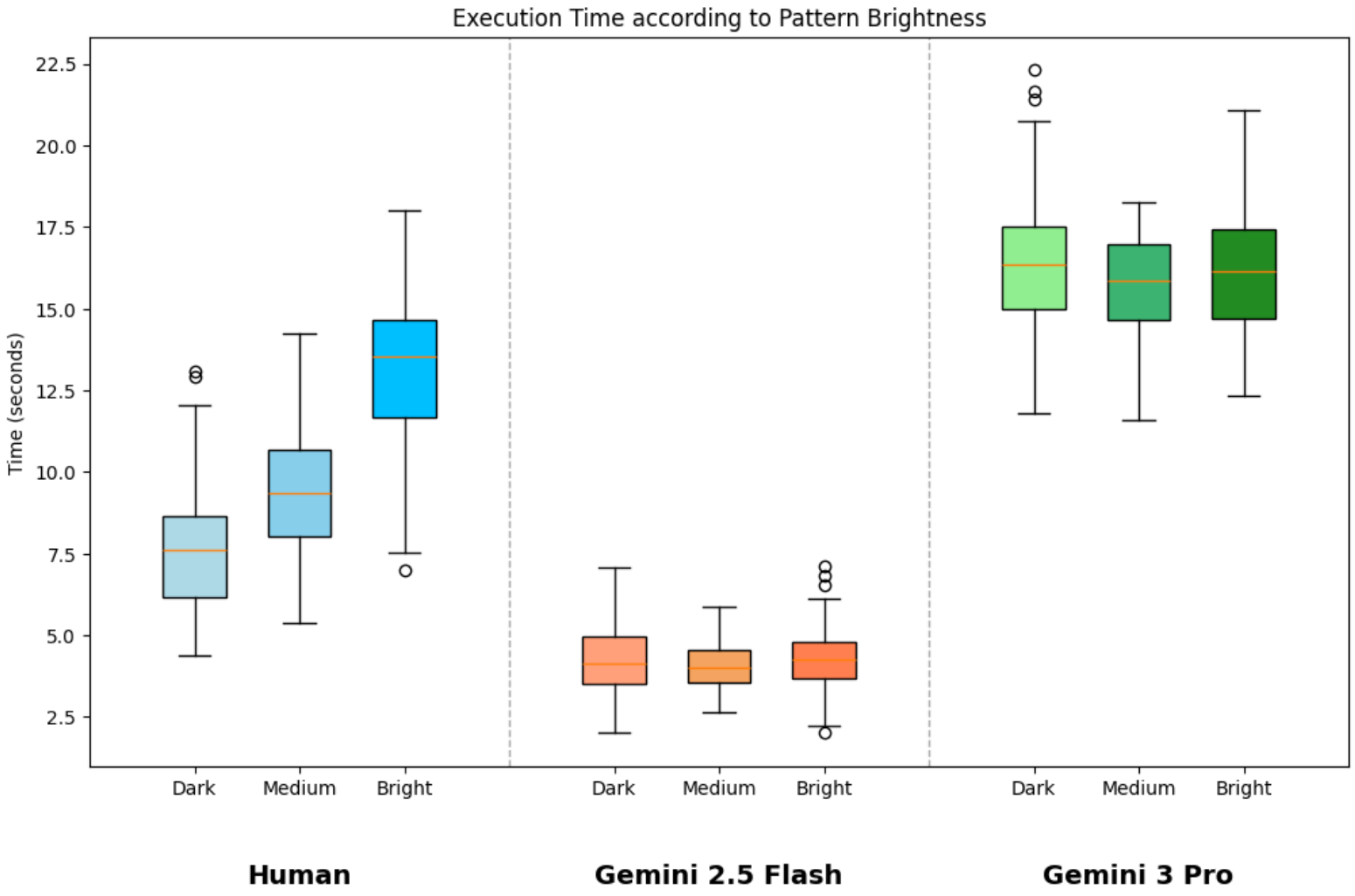
Evaluation


Execution Time according to Pattern Density
Time (seconds)
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Sparse
Medium
Dense
Sparse
Medium
Dense
Sparse
Medium
Dense
Human
Gemini 2.5 Flash
Gemini 3 Pro

Evaluation


Execution Time according to Pattern Shape
Time (seconds)
25
20
15
10
5
Hexagon
Square
Triangle
Hexagon
Square
Triangle
Hexagon
Square
Triangle
Human
Gemini 2.5 Flash
Gemini 3 Pro

Evaluation


Execution Time according to Pattern Brightness
Time (seconds)
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Dark
Medium
Bright
Dark
Medium
Bright
Dark
Medium
Bright
Human
Gemini 2.5 Flash
Gemini 3 Pro

Conclusion

- In this work, we experimentally demonstrate that changing the characteristics of the image significantly changes the user's CAPTCHA completion time, even when based on the same optical illusion
  - This trend is a clear distinction from AI, where execution time is almost constant when the image size is the same
  - Leveraging these characteristics, ILLUnatic is expected to show advantages in privacy protection over existing CAPTCHA systems by effectively distinguishing between normal users and bots based solely on execution time
- However, some experimental participants reported eye fatigue for certain patterns, so additional user evaluation may be necessary before practical application

Thank you